

# DESDEO2: Core-Logic for Implementing Interactive Multiobjective Optimization Methods

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## Software

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## Summary

Many real-world problems involve optimizing multiple conflicting objective functions simultaneously: when designing a car, for example, reducing its manufacturing cost typically comes at the expense of passenger safety and fuel efficiency. In such multiobjective optimization problems, there is no single optimal solution but, instead, multiple compromise solutions representing different trade-offs.

To select the best compromise, a decision maker (a domain expert) must explore the trade-offs and choose the solution they find most preferred. [Interactive methods](#) ([Miettinen, 1999](#); [Miettinen et al., 2016](#)) support this task. They proceed step by step, letting the decision maker examine candidate solutions, express preferences, and steer the search toward more desirable solutions, learning along the way about the problem and which preferences are achievable.

DESDEO2 is an open source software framework that provides a foundation for implementing interactive multiobjective optimization methods. It is a full redesign of the earlier DESDEO software ([Misitano et al., 2021](#)), developed to address practical challenges in building reusable and reproducible decision-support tools. We refer to the new version as DESDEO2 to distinguish it from its predecessor; outside this context, it is simply known as DESDEO. The scope of this paper is limited on the Python-based core-logic layer of DESDEO2, which is fully self-contained and [usable as a standalone library](#). Outside the core-logic, the broader framework also includes a web API and a graphical web user interface, both under active development.

The core-logic provides explicit abstractions for problem modeling, method execution, and preference handling, enabling interactive methods to be implemented as modular, composable components that are decoupled from any particular interface or deployment technology. This improves the reproducibility of experiments, reusability, and extensibility. The core-logic has been utilized in research and teaching, and is ready for broader use.

## Statement of need

Multiobjective optimization methods can be classified based on the timing when the preference information of a decision maker is used ([Miettinen, 1999](#)): *a priori* methods elicit preferences before optimization, *a posteriori* methods first generate a representative set of compromise solutions from which the decision maker selects one, while *interactive* methods incorporate preferences iteratively to generate new desirable solutions during the solution process. Because preferences unfold in an iterative dialogue with the decision maker, interactive methods are inherently software-intensive: any usable implementation must maintain state across iterations,

41 support diverse preference types, and let the decision maker drive the control flow. Providing  
42 these capabilities as reusable components, rather than re-implementing them for every method,  
43 is therefore valuable.

44 Existing open source frameworks do not explicitly provide reusable abstractions for these  
45 iterative, preference-driven workflows. Without such software, researchers must repeatedly  
46 reimplement not only optimization methods but also interaction logic, state management, and  
47 preference handling, making systematic experimentation laborious and difficult to reproduce  
48 (Afsar et al., 2021, 2024). DESDEO2's core-logic is designed to fill this gap. It targets  
49 researchers in interactive multiobjective optimization, who can experiment with existing  
50 methods and develop new ones; students learning about interactive methods; and practitioners  
51 building decision-support systems.

52 The need for reusable implementations originally motivated the development of earlier versions  
53 of DESDEO (Misitano et al., 2021; Ojalehto & Miettinen, 2018), which have enabled a range  
54 of successful research applications and practical decision-support, e.g., (Afsar, Ruiz, et al.,  
55 2023; Afsar, Silvennoinen, et al., 2023; Burkotová et al., 2024; Eyvindson et al., 2024; Kania et  
56 al., 2022, 2023). However, accumulated experience revealed architectural limitations: problem  
57 definitions were not clearly separated from method logic; method state, such as iteration  
58 history and preference information, was not explicitly modeled; and connectivity to external  
59 solvers and support for important problem types, such as mixed-integer and scenario-based  
60 problems, were limited.

## 61 State of the field

62 Many open source frameworks support research on non-preference-based, *a priori*, and *a*  
63 *posteriori* multiobjective optimization methods. Libraries, such as jMetal (Durillo et al., 2010),  
64 PlatEMO (Tian et al., 2017), pymoo (Blank & Deb, 2020), Platypus (Hadka, 2024), DEAP  
65 (Fortin et al., 2012), pagmo/pygmo (Biscani & Izzo, 2020), and ParMOO (Chang & Wild,  
66 2023), have become commonplace, particularly for evolutionary multiobjective optimization  
67 approaches; fewer software exist for scalarization-based methods, one notable exception being  
68 vOptSolver and its successor MultiObjectiveAlgorithms.jl (Dowson et al., 2025). These tools  
69 provide rich support for solution generation and benchmarking, and some offer preference-based  
70 mechanisms, such as reference point integration. However, these features are designed mainly  
71 for *a priori* use; support for the iterative, decision-maker-driven workflows that characterize  
72 interactive methods is lacking. Since existing frameworks assume in their core execution model  
73 a single optimization run without structured intervention, accommodating interactive workflows  
74 would require changes to core abstractions rather than incremental extensions, making it  
75 impractical to contribute these features to existing tools.

76 Without dedicated frameworks, implementations of interactive methods have typically been  
77 standalone prototypes or tightly coupled to specific applications, making them difficult to  
78 reuse, compare, or extend beyond their original context, e.g., (Siraj et al., 2015; Vetturini et  
79 al., 2024). Earlier versions of DESDEO (Misitano et al., 2021; Ojalehto et al., 2016) began  
80 addressing this lack by providing an open source framework specifically for interactive methods,  
81 and DESDEO2 builds on this foundation with its restructured and redesigned core-logic. Rather  
82 than replacing existing optimization frameworks, DESDEO2 complements them by allowing  
83 solvers from other libraries, such as SciPy (Virtanen et al., 2020), to be used as computational  
84 backends within its interactive method workflows.

## 85 Software design

86 A foundational design decision in DESDEO2 is to represent optimization problems in a  
87 serializable, language-agnostic form that can be defined once and then evaluated, stored, and  
88 exchanged across tools and software boundaries. Building on this, the core-logic addresses five

key challenges identified from experience with earlier versions of DESDEO: **C1** problem modeling, **C2** language-agnostic problem representation, **C3** interactive method state management, **C4** enabling extensions, and **C5** usage and contribution support.

To address C1 and C2, DESDEO2 represents problem definitions as explicit Pydantic<sup>1</sup> models in Python, which can be exported to and reconstructed from a [JSON representation](#). The model is designed for multiobjective optimization and supports [data-driven settings](#), where objective and constraint values may be computed by external simulations or opaque-box models. [Evaluators and parsers](#) bridge the model to external problem-definition ecosystems, and [solver interfaces](#) connect the core-logic to external optimizers; supported backends currently include Pyomo ([Hart et al., 2011](#)), SymPy ([Meurer et al., 2017](#)), CVXPY ([Diamond & Boyd, 2016](#)), Gurobi ([Gurobi Optimization, LLC, 2026](#)), Polars ([Vink et al., 2025](#)), SciPy ([Virtanen et al., 2020](#)), nevergrad ([Rapin & Teytaud, 2018](#)), and the COIN-OR solvers CBC ([Forrest et al., 2026](#)), Bonmin ([Bonami et al., 2008](#)), and Ipopt ([Wächter & Biegler, 2006](#)) (see the [documentation](#)). Problems thus need to be modeled only once to be [scalarized](#), evaluated, and solved in numerous ways (C1), while remaining storable and exchangeable across software boundaries, e.g., in databases ([Saini et al., 2023](#)) (C2).

Regarding state management (C3), earlier DESDEO versions coupled method state tightly to execution logic, making it difficult to persist, inspect, or reuse outside the running process. DESDEO2 instead provides explicit state representations with well-defined transitions, leaving persistence to the surrounding system, so interactive processes can be stored, resumed, and compared systematically, particularly when integrated with databases in decision-support systems; for standalone use, optional utility functions handle common state bookkeeping locally.

To enable extensions (C4), instead of implementing methods as monolithic algorithms, DESDEO2 decomposes functionality into reusable components for problem handling, preference processing, and solution generation, e.g., scalarization and [evolutionary operators](#). Individual components can be replaced, hybridized ([Sindhya et al., 2013](#)), or extended without re-implementing entire methods. Whereas the previous DESDEO was distributed as four separate packages, DESDEO2 is a single cohesive package whose [module structure](#) mirrors their high-level roles ([Misitano et al., 2021](#)) with fully redesigned contents: we deliberately prioritized a solid foundation for future development over backwards compatibility, supporting extensibility (C4) while keeping the code-base navigable (C5).

Finally, to support usage and contribution (C5), the [documentation](#) is inspired by the Diátaxis approach ([Procida, 2026](#)), separating tutorials, how-to guides, explanations, and reference material, and the core-logic is accompanied by unit tests targeting individual components. As a current limitation, DESDEO2's web-facing components, which would extend it into a full decision-support system, are not yet complete.

## Research impact statement

DESDEO has supported a sustained body of research on interactive multiobjective optimization over several years. Earlier versions enabled comparative studies, decision-support applications, and practical deployments across multiple domains, such as engineering design, forest management, and production planning (e.g., ([Afsar, Ruiz, et al., 2023](#); [Afsar, Silvennoinen, et al., 2023](#); [Burkotová et al., 2024](#); [Eyvindson et al., 2024](#); [Kania et al., 2022, 2023](#))), establishing both the community's need for reusable interactive method implementations and the research experience that informed DESDEO2's design.

DESDEO2's redesign enables lines of research that its predecessor's architecture could not readily support. Because interactive methods are now decoupled from any particular interface and share common abstractions for problems, preferences, and state, they can be implemented,

<sup>1</sup><https://github.com/pydantic/pydantic>, accessed 19 August 2026.

hybridized, and compared systematically rather than rebuilt as one-off prototypes. This makes it feasible to combine scalarization-based and evolutionary methods within a single framework, to switch methods or preference types during a solution process, and to store, resume, and compare interactive sessions, which supports reproducible studies of decision maker behavior, for example. The serializable problem representation similarly extends such research to previously impractical problem classes, including mixed-integer, scenario-based, and simulation/data-driven problems.

DESDEO2's core-logic is actively used in ongoing research at the University of Jyväskylä, supporting multiple Research Council of Finland funded projects and doctoral and master-level theses; recent examples include (Pajasmaa et al., 2026; Saini et al., 2025, 2026; Tahvanainen et al., 2026). The framework has been presented at multiple international conferences and used in university-level teaching, including an international summer school in 2025, and in public demonstrations during European Researchers' Night events. Since early 2024, DESDEO2's development branches have accumulated over 2000 commits. Development is led by the Multiobjective Optimization Group (<https://optgroup.it.jyu.fi/>) at the University of Jyväskylä, and contributions from the broader research community are warmly welcomed.

## AI usage disclosure

Generative tools based on large language models have been used to partly support the development of DESDEO2: for code snippet generation, refactoring, exploring the documentation and code bases of third-party libraries, and writing documentation, such as parameter descriptions. All generated code and documentation has been audited and validated by a human. Generative tools were also used to plan the initial structure of this paper based on the authors' earlier draft. All the final contents and ideas in this paper have been produced and verified by the authors.

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